eters in the paint shop using different probe connections. Along with a hot wire probe, it measures and restricts desired air velocity. Testo 440 with IAQ probe maintains the air quality of the workspace.

"Testo dew point transmitters regulate and monitor dew content in the sprayed air in paint shop applications.

"Testo compressed air counters determine and avoid any kind of air leakages in the process.

"Testo humidity and temperature transmitters monitor critical climate during process engineering. These allow the operator or engineer to maintain the necessary conditions.

"Testo flue gas analyser, Testo 320 and Testo 330 measure the efficiency of heating dryers in the baking process.

"Thermal imagers check cold air leakage from windows and doors, and monitor the thermal profile of tyres in wear-and-tear testing, temperature of brakes, exhaust silencer and engine (with complete thermal profile of the engine)."

Software platforms for automotive testing

FlexLogger is a configuration-based, data-logging software that targets ad-hoc measurements for validation of physical systems prevalent throughout automotive sub-systems. The software is designed to configure and execute measurement and logging tasks without programming.

CompactRIO with DAQmx is a measurement and control platform that combines the powerful, reconfigurable field programmable gate array (FPGA) of CompactRIO with DAQmx driver API for host side of the application. It combines the best of CompactDAQ and CompactRIO, while upgrading the controllers with time-sensitive networking capabilities for network determination and reliability.

Jeff Phillips, head - automotive marketing, National Instrument, says, "NI's open software-centric platform provides flexibility and customisations necessary to meet current and future automotive test needs, while maintaining the legacy of test systems and reducing development time. It can take any measurement on any network bus, and combine electrical measurements with physical ones to validate a system of systems, such as adaptive cruise control or lane-keep assist.

"The software-defined platform is a collection of highly interoperable software and hardware products that meet the spectrum of needs from custom development environments to configuration-based software, and plug-in data acquisition boards to high-performance reconfigurable FPGAs for advanced processing and control. NI serves across the entire design cycle, from design and characterisation, through software and hardware validation, to mechanical and end-of-line testing."

Challenges in T&M for automobiles

Evolution of automotive infotainment systems requires a combination of car radio, car navigation system, digital television, Internet modem, smartphone and more. Hardware supporting completely different standards may be designed on the same chip or on multiple chips assembled side by side.

As the spectrum is limited, many of these standards operate quite close to each other. Communication standards have significant proximity and may overlap in spectrum.

Adjacent channel interference is another potential problem that needs to be considered. A wanted signal is degraded because of poor power control and/or inadequate transmitter or receiver filtering.

A new era of mobility is upon us. Critical technology trends in ADAS, electric vehicles and V2X are bringing new test challenges and pressures beyond the here and now. The quickly evolving technology landscape increases the pressure to any test schedule and requirement. There is a need to ensure that test managers are confident in their ability to overcome the pressures of rapidly changing test requirements, using an open and easily upgradable platform designed for test system openness and flexibility. EFY